

Notes: Cornelli & Goldreich (JF, 2003)

Bookbuilding: How Informative Is the Order Book?

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1 Big picture

- **Question.** Is bookbuilding primarily a distribution mechanism, or does it actually extract pricing information from institutional investors?
- **Setting.** The paper studies 63 international equity issues run by a major European investment bank between 1995 and 1999: 37 IPOs and 26 seasoned equity offerings (SEOs).
- **Unique data.** The paper observes the actual institutional order books: bid quantities, bid timing, bid types, limit prices, bidder frequency, allocations, and issue outcomes.
- **Main result.** The issue price responds strongly to information in the book. The quantity-weighted average limit price explains most cross-sectional variation in normalized IPO offer prices.
- **Bid type result.** Limit and step bids contain the most useful pricing information. Strike bids mainly contribute to aggregate oversubscription.
- **Bidder identity result.** Large bidders, frequent bidders, and bidders favored in allocation have more influence on the issue price, consistent with the idea that informed investors are rewarded for revealing information.
- **Public versus private information result.** Public information affects issue prices largely through investor bids, not directly. This blurs the line between public and private information in bookbuilding.
- **Aftermarket result.** Oversubscription and demand elasticity contain residual information not fully impounded into the issue price, and help predict first-day aftermarket performance and volatility.
- **SEO result.** Even when an observable premarket price exists, underwriters use order-book limit prices to set the SEO issue price.
- **Economic takeaway.** Bookbuilding is an information-production mechanism. Investors reveal information in exchange for favorable allocations, and underwriters selectively use that information in pricing.

2 Incremental contribution of the paper

- **Direct order-book evidence.** Prior work inferred bookbuilding behavior from offer prices, price revisions, allocations, or aftermarket returns. This paper observes the book itself, allowing direct measurement of what investors reveal before the issue price is set.
- **Separation of price and quantity information.** The paper distinguishes limit prices, step bids, strike bids, oversubscription, and elasticity. This is an incremental advance over studies that use only price-range revisions or final underpricing.
- **Investor-type information.** By distinguishing large, small, frequent, infrequent, favored, nonfavored, early, and late bidders, the paper connects information production to investor relationships and allocation favoritism.
- **Public-information mediation.** The paper contributes to the partial-adjustment debate by showing that market returns, high-tech status, and IPO activity affect issue prices mostly to the extent that they are reflected in investor bids.
- **Aftermarket validation.** The paper does not merely show that bids predict offer prices. It asks whether bid information is correct by examining first-day returns and volatility.
- **IPO and SEO comparison.** Most bookbuilding theories focus on IPOs. The paper shows that bookbuilding also matters in SEOs, where an existing market price is available but may be illiquid or noisy.
- **Mechanism-level contribution.** The evidence directly supports the Benveniste–Spindt view that underpricing and allocation can compensate investors for revealing information.

3 Data and measurement

3.1 Sample of issues

The data come from the institutional order books of a major European investment bank. The sample contains 63 international equity issues from 1995 to 1999:

- 37 initial public offerings (IPOs);
- 26 seasoned equity offerings (SEOs);
- 20 privatizations across IPOs and SEOs;
- issuing firms from 24 countries and multiple industries.

The bank was global bookrunner for the issues and had a major Wall Street presence, making it comparable to large U.S. banks in bookbuilding practice.

Interpretation: The key value of the sample is not size but depth. The authors see the actual bidding behavior before pricing, which is normally confidential.

3.2 Bookbuilding procedure

For IPOs, the underwriter announces an indicative price range before collecting bids. Institutional investors then submit bids over roughly two weeks. After the book closes, the underwriter and issuer choose the issue price.

For SEOs, there is no preliminary price range because the existing market price serves as a preissue benchmark. The offer price is usually set as a discount or premium relative to the last premarket price.

Interpretation: The underwriter has discretion: the issue price is not automatically set where demand equals supply. This discretion is exactly what allows the paper to ask whether the book is used for information extraction.

3.3 Bid types

The paper distinguishes three bid types:

- **Strike bids:** a demand for shares at whatever final issue price is set.
- **Limit bids:** a demand for shares up to a maximum price.
- **Step bids:** a demand schedule with different quantities at different prices; conceptually, a bundle of limit bids.

Interpretation: Limit and step bids reveal price-sensitive information about valuation. Strike bids mostly reveal quantity demand, not price willingness.

3.4 Issue-price normalization

For IPOs, the issue price is normalized relative to the initial price range:

$$\text{Normalized issue price} = \frac{P_I - P_{\min}}{P_{\max} - P_{\min}}, \quad (1)$$

where P_I is the final issue price and $P_{\min}$ and $P_{\max}$ are the lower and upper bounds of the initial range. Values below zero or above one indicate pricing outside the initial range.

For SEOs, prices are normalized relative to the last premarket price.

Interpretation: Normalization controls for the information already embedded in the preliminary price range. The regressions ask whether the book explains where inside or outside that range the issue is priced.

3.5 Order-book variables

The main book variables are:

- **Average limit price:** quantity-weighted average of all limit prices, normalized by the initial range for IPOs or by the premarket price for SEOs.
- **Oversubscription:** the logarithm of one plus total demand divided by supply.
- **Elasticity:** elasticity of demand from the issue price or average limit price to a price one percent higher.
- **Range size:** initial price range divided by its midpoint, a proxy for ex ante uncertainty.
- **Bid timing:** early bids versus bids submitted in the final three days.
- **Bidder type:** large versus small, frequent versus infrequent, favored versus nonfavored, early versus late.

Interpretation: Average limit price measures valuation information; oversubscription measures quantity pressure; elasticity measures consensus among price-sensitive bidders.

3.6 Aftermarket variables

The main aftermarket variable is the first-day aftermarket return, benchmarked relative to the domestic stock market index. For SEOs, the paper also studies whether the aftermarket price reverts toward the premarket price.

Interpretation: Aftermarket tests ask whether the book contains information not fully incorporated into the issue price.

4 Empirical specifications

4.1 Issue price and order-book information

The baseline IPO pricing regression can be written as:

$$\text{IssuePrice}_j^{\text{norm}} = \alpha + \beta_1 \text{AvgLimitPrice}_j + \beta_2 \text{Oversubscription}_j + \beta_3 X_j + \varepsilon_j. \quad (2)$$

- The dependent variable is the final issue price normalized by the initial range.
- The main independent variables are the average limit price and oversubscription.
- Additional specifications add elasticity, range size, privatization interactions, and variants of the dependent variable.

Interpretation: If the bank only uses the book for distribution, the coefficients on bid variables should be weak. Strong coefficients indicate information extraction.

4.2 Bidder-type regressions

The paper estimates versions of the pricing regression separately by bidder type:

$$\text{IssuePrice}_j^{\text{norm}} = \alpha + \beta_H \text{AvgLimitPrice}_{j,H} + \beta_L \text{AvgLimitPrice}_{j,L} + \gamma_H \text{Oversubscription}_{j,H} + \gamma_L \text{Oversubscription}_{j,L} + \varepsilon_j. \quad (3)$$

The categories include large/small bids, frequent/infrequent bidders, favored/nonfavored bidders, and early/late bids.

Interpretation: These regressions test whether the underwriter distinguishes informed from less-informed demand. The strongest effects come from the same investors who are favored in allocations.

4.3 Public information and bid mediation

The public-information tests use variables such as market index returns, high-tech industry status, firm size, and industry IPO activity. The logic is:

1. Does public information move the issue price?
2. Does public information move bids?
3. Once bids are included, does public information still directly move the issue price?

Interpretation: This mediation structure is central. The paper argues that public information matters mainly through investors' interpretation of it, as expressed in bids.

4.4 Aftermarket return regressions

For IPOs, the first-day return regression is:

$$R_j^{1d} = \alpha + \beta_1 \text{IssuePrice}_j^{\text{norm}} + \beta_2 \text{Oversubscription}_j + \beta_3 \text{AvgLimitGap}_j + \beta_4 \text{Elasticity}_j + \varepsilon_j. \quad (4)$$

For SEOs, the regressions include the percentage difference between the premarket price and issue price:

$$R_j^{1d} = \alpha + \beta(P_j^{\text{pre}} - P_j^I)/P_j^I + \Gamma'Z_j + \varepsilon_j. \quad (5)$$

Interpretation: Aftermarket returns show whether the underwriter fully or only partially incorporates order-book information.

5 Identification strategy

5.1 What is identified?

The paper identifies how the underwriter uses order-book information in pricing. This is not a randomized experiment; it is a direct observational test using confidential book data. The key identifying idea is that the initial price range captures pre-bookbuilding information, while within-range revisions reflect information collected during bookbuilding.

5.2 Why normalization matters

Normalizing by the initial price range controls for information available before bidding. If the underwriter already knew everything relevant, bids should not explain where within the range the price is set. If bids explain the normalized issue price, they contain incremental information.

5.3 Bid type and bidder type as mechanism tests

The paper does not rely only on an aggregate correlation between demand and price. It separately tests whether the most informative bids are:

- price-sensitive limit and step bids;
- large bids;
- bids from frequent investors;
- bids that receive favored allocations;
- late bids that can incorporate recent information.

Interpretation: The pattern matters more than any single coefficient. It shows that the bank reacts to information-rich parts of the book, not simply to noisy demand.

5.4 Public information tests as identification reinforcement

Public information can challenge the information-extraction interpretation because partial adjustment to public information is not normally necessary for incentive compatibility. The paper shows that public information enters bids and that issue prices respond to the bids, not necessarily directly to the public signal.

Interpretation: This helps reconcile public-information partial adjustment with bookbuilding theory: public signals still need investor interpretation to become useful pricing information.

5.5 Aftermarket validation

If the book contains real information, order-book variables should predict aftermarket outcomes when they are not fully incorporated into the issue price. The paper finds exactly this for oversubscription and elasticity.

Interpretation: This is a validation step. It asks whether the book variables that move offer prices and residual returns are economically meaningful, not merely correlated with underwriter choices.

5.6 Limitations

- The data come from one investment bank, though the paper argues it is comparable to large U.S. banks.
- The sample is small: 37 IPOs and 26 SEOs.
- The underwriter's objective function is not observed directly.
- Bidder information and strategic behavior are not experimentally separated.
- Favorable allocation is observed ex post, so allocation-favored bid regressions are mechanism evidence rather than causal proof.

Interpretation: The paper's strength is direct measurement of confidential book data, not large-sample random variation.

6 Tables and figures

6.1 Table I: Summary Statistics of Issue and Bid Characteristics

Read:

- The sample contains 37 IPOs and 26 SEOs, including 20 privatizations.
- IPO average first-day benchmarked return is 7.57%; SEO average first-day benchmarked return is 3.27%.
- IPOs receive many bids: the average issue has 411 bids and 62 limit/step bids.
- Average oversubscription is 9.1 at the issue price for IPOs and 3.1 for SEOs.
- Limit and step bids are a minority of bids but contain rich price information.

Interpretation: The summary statistics show why bookbuilding can reveal information: the bank observes hundreds of institutional bids, including price-sensitive demand schedules, before setting the price.

Economic message: Even though most bids are strike bids, the smaller set of limit and step bids can determine how the underwriter interprets demand.

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Table I
Summary Statistics of Issue and Bid Characteristics

This table reports summary statistics of issue and bid characteristics for the IPOs and SEOs in the sample. All statistics (except *Number of issues* and *Number of privatizations*) are computed across the issues. *Issue price* and *Average limit price* (the quantity-weighted average of all limit prices in an issue) are normalized by the initial price range for IPOs and by the last premarket price for SEOs. *First-day return* is benchmarked relative to the domestic stock market index. *Elasticity* is the elasticity of demand computed from the issue price or the average limit price to a price one percent higher. *Oversubscription* is the total demand for shares divided by the total number of shares issued (including overallocation).

		IPOs	SEOs
Number of issues		37	26
Number of privatizations		8	12
Range size (relative to midpoint)	<i>mean</i>	16.3%	N/A
	<i>median</i>	14.8%	N/A
Issue price (normalized)	<i>mean</i>	0.51	-2.17%
	<i>median</i>	0.67	-1.92%
	<i>max</i>	1.50	0.26%
	<i>min</i>	-1.20 (relative to range)	-7.24% (relative to premarket price)
First-day return (benchmarked)	<i>mean</i>	7.57%	3.27%
	<i>median</i>	4.29%	3.73%
Number of bids per issue	<i>mean</i>	411	236
	<i>median</i>	375	172
Number of limit & step bids per issue	<i>mean</i>	62	55
	<i>median</i>	29	41
Average limit price (normalized)	<i>mean</i>	0.49	-1.48%
	<i>median</i>	0.49	-1.61%
	<i>max</i>	1.55	2.73%
	<i>min</i>	-0.92 (relative to range)	-9.57% (relative to premarket price)
Standard deviation of limit prices (normalized)	<i>mean</i>	0.75%	0.36%
	<i>median</i>	0.48% (relative to range midpoint)	0.27% (relative to premarket price)
Elasticity			
at issue price	<i>mean</i>	12.8	17.5
	<i>median</i>	3.6	13.7
at average limit price	<i>mean</i>	6.4	16.8
	<i>median</i>	1.0	10.2
Oversubscription			
at issue price	<i>mean</i>	9.1	3.1
	<i>median</i>	4.5	2.8
	<i>max</i>	62.3	7.9
	<i>min</i>	1.2	1.3
all bids	<i>mean</i>	9.9	3.4
	<i>median</i>	5.4	3.1
strike bids	<i>mean</i>	8.0	2.1
	<i>median</i>	3.0	2.1
limit and step bids	<i>mean</i>	1.9	1.3
	<i>median</i>	1.4	1.2

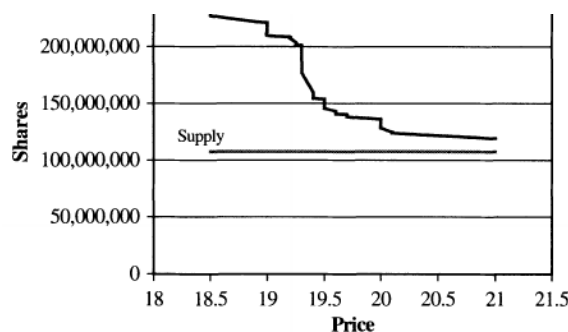


Figure 1: Example of supply and demand curves. This figure shows the supply and demand curves for a single issue in our sample. In this example, demand is relatively inelastic at higher prices and becomes more elastic as the price decreases.

6.2 Figure 1: Example of Supply and Demand Curves

Read:

- The figure shows one issue's book demand curve and the supply of shares.
- Demand is above supply throughout the price range.
- The underwriter still does not mechanically set the price at market clearing.

Interpretation: The figure makes visible the discretion in bookbuilding. Even when demand exceeds supply, the bank sets the issue price based on judgment, not an auction-clearing rule.

Economic message: Bookbuilding is not a standard auction. It is an information-gathering and discretionary pricing process.

6.3 Table II: The Choice of the Issue Price (IPOs)

Read:

- The average limit price alone explains 81.5% of the cross-sectional variation in normalized IPO prices.
- The coefficient on average limit price is close to one, so issue prices move almost one-for-one with price-sensitive bids.
- Oversubscription is statistically significant but less informative than average limit prices.
- Excluding range endpoints or range-interior-only issues does not remove the main result.

Interpretation: This is the main pricing table. It shows that the underwriter uses price-sensitive order-book information rather than relying only on total demand or the preliminary range.

Economic message: Limit prices reveal institutional investors' valuation information, and the bank uses that information in pricing.

Table II
The Choice of the Issue Price (IPOs)

This table reports regression coefficients for various model specifications. Heteroskedasticity-adjusted *t*-statistics, using White's (1980) variance-covariance matrix, are in parentheses. Except where otherwise specified, the dependent variable is the issue price normalized by the initial price range. *Average limit price* is the quantity-weighted average of all limit prices and is normalized by the initial price range. *Oversubscription* is the logarithm of $1 + \text{total demand/supply of shares}$, with demand measured at various prices. *Elasticity* is the elasticity of the logarithm of demand computed from the average limit price to a price one percent higher. *Range size* is measured relative to its midpoint. In Regression 30, the dependent variable is the issue price divided by the range midpoint, and the average limit price is normalized by the range midpoint. In Regression 11, the dependent variable is the absolute value of (issue price/average limit price - 1).

Dependent Variable (Regress 1 to 9): Issue Price (Normalized by Initial Price Range)	Regress 1	Regress 2	Regress 3	Regress 4	Regress 5	Regress 6	Regress 7	Regress 8	Regress 9	Regress 10	Regress 11
Intercept	-0.06 (-0.09)	-0.31 (-1.3)	-0.34 (-4.0)	-0.24 (-3.6)	-0.12 (-1.1)	-0.25 (-1.8)	-0.35 (-4.0)	-0.30 (-3.5)	-0.22 (-3.3)	-0.35 (-4.3)	0.04 (1.9)
Average limit price	1.12 (13.3)	0.96 (11.0)	0.97 (10.2)	1.30 (13.0)	0.93 (9.1)	1.05 (10.3)	0.90 (6.7)	0.91 (6.7)	0.90 (6.7)	0.91 (6.7)	-0.01 (-1.0)
Oversubscription (all bids)		0.41 (4.5)	0.18 (4.0)								
Oversubscription (strike bids)				0.15 (3.9)							
Oversubscription (limit and step bids)					0.07 (0.8)						
Elasticity						0.01 (0.4)					-0.01 (-2.4)
Range size							-0.68 (-1.3)			0.07 (0.8)	
Average limit price (pretentions only)								0.92 (7.9)			
Average limit price (nonpretentions only)									0.97 (10.3)		
Adjusted <i>R</i> -squared	81.5%	37.5%	87.3%	87.4%	81.3%	86.9%	87.0%	91.7%	81.9%	88.8%	2.2%
<i>N</i>	35	37	36	35	35	35	35	23	18	35	35

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Table III
Influence of Different Types of Bidders (IPOs)

This table reports regression coefficients (and heteroskedasticity-adjusted *t*-statistics in parentheses) for various model specifications. The dependent variable is the issue price normalized by the initial price range. *Average limit price* is the quantity-weighted average of all limit prices and is normalized by the initial price range. *Oversubscription* is the logarithm of $1 + \text{total demand/supply of shares}$. Large (small) bids are bids with a quantity above (below) the median in an issue. Frequent (infrequent) bidders are bidders who participate in at least (fewer than) three issues. *Favored* bids are bids that are awarded more shares (as a percentage of the bid quantity) than the median bid. The symbols ^{***}, ^{**}, ^{*} denote pairs of coefficients that are significantly different from each other at the one and five percent confidence levels, respectively.

Panel A: Average Limit Price of Different Types of Bidders

Dependent Variable: Issue Price (Normalized by Initial Price Range)	Regress 1	Regress 2	Regress 3
Intercept	-0.32 (-4.7)	-0.36 (-3.1)	-0.41 (-5.1)
Oversubscription (all bids)	0.18 (4.3)	0.15 (2.6)	0.18 (4.3)
Average limit price (large bids)	1.14 ^{***} (6.2)		
Average limit price (small bids)	-0.20 ^{***} (-1.2)		
Average limit price (frequent bidders)		1.25 ^{***} (4.8)	
Average limit price (infrequent bidders)		-0.29 ^{***} (-1.3)	
Average limit price (favored bids)			1.15 ^{***} (9.0)
Average limit price (nonfavored bids)			-0.22 ^{**} (-1.9)
Adjusted <i>R</i> -squared	86.7%	86.0%	89.1%
<i>N</i>	34	29	32

Panel B: Oversubscription of Different Types of Bidders

Dependent Variable: Issue Price (Normalized by Initial Price Range)	Regress 4	Regress 5	Regress 6
Intercept	-0.38 (-3.0)	-0.29 (-2.7)	-0.27 (-2.6)
Average limit price (all bids)	0.06 (1.3)	0.08 (0.9)	0.09 (1.1)
Average limit price (large bids)	0.34 (2.7)		
Oversubscription (small bids)	-0.13 (-0.7)		
Oversubscription (frequent bidders)		0.15 (2.5)	
Oversubscription (infrequent bidders)		0.05 (0.5)	
Oversubscription (favored bids)			0.01 (0.2)
Oversubscription (nonfavored bids)			0.25 (2.9)
Adjusted <i>R</i> -squared	87.0%	86.3%	84.9%
<i>N</i>	35	35	35

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6.4 Table III: Influence of Different Types of Bidders (IPOs)

Read:

- Average limit prices from large bids matter more than those from small bids.
- Average limit prices from frequent bidders matter more than those from infrequent bidders.
- Bids favored in allocation are more influential than nonfavored bids.
- Oversubscription from large and frequent bidders predicts issue price more strongly than oversubscription from small and infrequent bidders.

Interpretation: The table shows that the bank does not treat all demand equally. It relies more on bids likely to contain informed valuation signals.

Economic message: The results support the information-rent view of bookbuilding: investors reveal information and receive favorable allocations.

6.5 Table IV: The Evolution of the Book: How Early Bids Influence the IPO Price

Read:

- Early limit bids already contain pricing information.
- Late bids subsume much of the early information when both are included.
- Changes in the average limit price over the bookbuilding period are strongly related to the issue price.
- Late oversubscription is more informative than early oversubscription.

Interpretation: Information arrives throughout bookbuilding. The underwriter learns early but updates as bids are revised and new bids arrive near the end.

Economic message: Dynamic order-book updating is part of the information-production process.

The Evolution of the Book: How Early Bids Influence the IPO Price

This table reports regression coefficients (and heteroskedasticity-adjusted *t*-statistics in parentheses) for various model specifications. The dependent variable is the issue price normalized by the initial price range. *Average limit price* is the quantity-weighted average of all limit prices and is normalized by the initial price range. *Oversubscription* is the logarithm of 1 + total demand/supply of shares. Early bids are bids received at least three days prior to the closing of the book. Late bids are bids received in the last three days. Δ *Avg limit price* is the difference between the average limit price in the final book and the average limit price from early bids.

Dependent Variable: Issue Price (Normalized by Initial Price Range)	Reg 1	Reg 2	Reg 3
Intercept	-0.25 (-3.1)	-0.23 (-3.0)	-0.28 (-3.9)
Average limit price (all bids)			0.96 (10.7)
Average limit price (early bids)	0.90 (10.3)	-0.64 (-2.6)	
Average limit price (late bids)		1.57 (6.4)	
Δ Avg limit price	1.32 (5.5)		
Oversubscription (all bids)	0.14 (2.6)	0.14 (3.5)	
Oversubscription (early bids)			0.07 (1.37)
Oversubscription (late bids)			0.14 (3.06)
Adjusted <i>R</i> -squared	87.0%	87.9%	86.2%
<i>N</i>	34	33	34

Table V
Public and Private Information – Market Index Return

This table reports coefficients (and heteroskedasticity-adjusted *t*-statistics in parentheses) for regressions related to the market index return during the bookbuilding period for IPOs. *Market return* is the return on the domestic stock market index over the bookbuilding period. *Average limit price* is the quantity-weighted average of all limit prices. The issue price and average limit price are normalized by the midpoint of the range. Early bids are bids submitted more than three days before the close of the book. Δ *Avg limit price* is the difference between the average limit price in the final book and the average limit price from early bids. In Regression 2, the independent variable *Portion of Δ avg limit price not explained by Market return* is the residual from Regression 1. In Regression 4, the dependent variable *Portion of Issue price not explained by Avg Limit Price* is the residual from Regression 3. The coefficients of the intercepts are not reported.

	Reg 1	Reg 2	Reg 3	Reg 4
Dependent Variables:	Δ Avg Limit Price	Issue Price (normalized by range midpoint)	Issue Price (normalized by range midpoint)	Portion of Issue Price Not Explained by Avg Limit Price
Independent Variables:	Market return 0.190 (2.24)	Average limit price (early bids) 0.077 (13.89)	Average limit price (early bids) 0.082 (13.84)	Market return 0.077 (0.84)
		Market return 0.372 (4.03)	Δ Avg limit price 1.585 (8.71)	
		Portion of Δ Avg limit price not explained by Market return 1.533 (8.52)		
	Adjusted <i>R</i> -squared = 17% <i>N</i> = 34	Adjusted <i>R</i> -squared = 82.8% <i>N</i> = 34	Adjusted <i>R</i> -squared = 82.7% <i>N</i> = 34	Adjusted <i>R</i> -squared = -2.2% <i>N</i> = 34

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6.6 Table V: Public and Private Information – Market Index Return

Read:

- Market returns during bookbuilding affect average limit prices and issue prices.
- Once the bid response is included, direct market-return effects are reduced.
- Residual changes in bids, not explained by market returns, still affect issue prices.

Interpretation: The table suggests that public market information is filtered through investor bidding. The underwriter needs to see how investors interpret public information.

Economic message: Public information can become economically relevant through private order-book responses.

6.7 Table VI: Public and Private Information – Firm Characteristics

Read:

- High-tech status, firm size, and industry IPO activity affect bids and sometimes issue prices.
- Industry IPO activity raises average limit prices and issue prices.
- Once bids are included, public firm-characteristic effects on issue price are weaker.

Interpretation: Public information such as industry conditions matters mostly because investors react to it in their bids. The bank uses the bid reaction rather than mechanically pricing off public variables.

Economic message: The distinction between public and private information is blurred in bookbuilding because public signals require investor interpretation.

Table VI
Public and Private Information – Firm Characteristics

This table reports coefficients (and heteroskedasticity-adjusted *t*-statistics in parentheses) for regressions related to the effect of firm characteristics, including *Industry*, *Size*, and *Industry IPO activity* on the *issue price* and on the *bids*. *High tech* is a dummy equal to one for issuers in a high-tech industry. *Size* is the logarithm of annual sales. *Industry IPO activity* is the number of worldwide IPOs in the industry of the issuer in the current month divided by the average number of monthly IPOs in the industry. The *issue price* and the *average limit price* are normalized by the midpoint of the range. *Oversubscription* is the logarithm of $1 + \text{total demand/supply of shares}$. The dependent variable *Portion of issue price not explained by the limit prices and Oversubscription* is the residual from Regression 10 of Table II. The coefficients of the intercepts are not reported.

Dependent Variable:	Issue Price (Normalized by Range Midpoint)	Average Limit Price	Oversubscription	Portion of Issue Price Not Explained by the Limit Prices and Oversubscription
	Reg 1	Reg 2	Reg 3	Reg 4
Independent Variable:	High tech 0.084 (0.45)	High tech 0.069 (2.28)	High tech 1.073 (2.88)	High tech 0.022 (4.03)
	Adj <i>R</i> ² = 5.9% <i>N</i> = 37	Adj <i>R</i> ² = 4.5% <i>N</i> = 35	Adj <i>R</i> ² = 14.7% <i>N</i> = 37	Adj <i>R</i> ² = 18.2% <i>N</i> = 35
	Reg 5	Reg 6	Reg 7	Reg 8
Independent Variable:	Size -0.011 (-2.19)	Size -0.008 (-0.92)	Size -0.064 (-0.92)	Size 0.188 (0.16)
	Adj <i>R</i> ² = 3.6% <i>N</i> = 36	Adj <i>R</i> ² = 1.5% <i>N</i> = 34	Adj <i>R</i> ² = -0.3% <i>N</i> = 36	Adj <i>R</i> ² = 3.6% <i>N</i> = 37
	Reg 9	Reg 10	Reg 11	Reg 12
Independent Variable:	Industry IPO activity 0.041 (4.60)	Industry IPO activity 0.022 (4.03)	Industry IPO activity 0.188 (1.16)	Industry IPO activity 0.055 (0.40)
	Adj <i>R</i> ² = 20.0% <i>N</i> = 37	Adj <i>R</i> ² = 18.2% <i>N</i> = 35	Adj <i>R</i> ² = 5.0% <i>N</i> = 37	Adj <i>R</i> ² = 0.2% <i>N</i> = 35

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Table VII
Aftermarket Return: IPOs

This table reports regression coefficients (and heteroskedasticity-adjusted *t*-statistics in parentheses) for various model specifications. The dependent variable is the first-day aftermarket return benchmarked relative to the domestic stock market index. *Issue price* is normalized by the initial price range. *Oversubscription* is the logarithm of $1 + \text{total demand/supply of shares}$. The average limit price is the quantity-weighted average of all limit prices. *Elasticity* is computed from the issue price (or average limit price) to a price one percent higher.

Dependent Variable: First-Day Return (Benchmarked)	Only Range Interior		Only Low Oversub- scription		Reg 5	Reg 6	Reg 7
	Reg 1	Reg 2	Reg 3	Reg 4	Reg 5	Reg 6	Reg 7
Intercept	0.02 (1.5)	-0.06 (-2.3)	-0.05 (-1.5)	-0.08 (-2.0)	-0.05 (-1.5)	-0.10 (-3.0)	-0.09 (-2.9)
Issue price (normalized by initial price range)	0.07 (3.6)	0.02 (1.1)					
Oversubscription (all bids)		0.06 (3.4)	0.06 (3.4)	0.08 (2.4)	0.06 (3.4)	0.08 (4.6)	0.07 (4.5)
Percentage difference between average limit price and issue price					-0.46 (-1.1)		
Elasticity (at issue price)						0.014 (2.4)	
Elasticity (at average limit price)							0.019 (1.4)
Adjusted <i>R</i> -squared	18.7%	32.5%	21.3%	13.5%	36.6%	35.3%	35.9%
<i>N</i>	36	36	19	22	34	36	34

measured by a low elasticity, means that investors have diverse opinions about the value of the shares and this diversity of opinions results in lower returns. This result is consistent with Kandel, Sarig, and Wohl (1999), who find that the demand elasticity in IPO auctions is positively related to aftermarket returns. They argue that elasticity affects the price of the shares for two reasons: first, an elastic demand may reflect more

6.8 Table VII: Aftermarket Return: IPOs

Read:

- Issues priced higher within the range have higher first-day returns.
- Oversubscription is positively related to aftermarket returns in several specifications.
- Elasticity measured at the issue price is positively related to aftermarket return.
- The relation implies partial adjustment: not all information in the book is incorporated into the offer price.

Interpretation: This table validates that book variables contain real valuation information. Some of that information remains in first-day returns because underwriters do not fully adjust the price.

Economic message: Information extraction and underpricing coexist: the underwriter learns from investors but leaves rents to compensate information revelation.

6.9 Table VIII: Effect of Uncertainty on Aftermarket Return and Volatility (IPOs)

Read:

- Lower elasticity, meaning less consensus among bidders, is associated with less predictable aftermarket returns.
- Elasticity is negatively related to aftermarket volatility over one-, two-, and four-week windows.
- Range size and elasticity capture different types of uncertainty: ex ante valuation uncertainty versus order-book disagreement.

Interpretation: Elasticity measures consensus among limit bidders. When bidders disagree more, aftermarket prices are more volatile and less predictable.

Economic message: Demand elasticity is an information-quality variable, not just a demand-quantity variable.

Table VIII
Effect of Uncertainty on Aftermarket Return and Volatility (IPOs)

This table reports regression coefficients (and heteroskedasticity-adjusted *t*-statistics in parentheses) for various model specifications. The dependent variable in Regressions 1 and 2 is the absolute value of the difference between the first-day return and the expected return, where the expected return is computed using the coefficient estimates in Regression 2 in Table VII. The dependent variables in Regressions 3 through 8 are the standard deviations of daily returns in the aftermarket over different time periods. Elasticity is computed from the issue price (or average limit price) to a price one percent higher. The average limit price is the quantity-weighted average of all limit prices. Range size is measured relative to its midpoint.

Dependent Variables:	Absolute Value of Unexpected First-day Return		Aftermarket Volatility							
			One-week Volatility		Two-week Volatility		Four-week Volatility			
	Reg 1	Reg 2	Reg 3	Reg 4	Reg 5	Reg 6	Reg 7	Reg 8		
Intercept	0.07 (8.1)	0.07 (-3.2)	0.17 (1.9)	0.16 (1.9)	0.20 (3.4)	0.22 (3.6)	0.14 (1.6)	0.12 (1.3)		
Elasticity (at the issue price)	-0.01 (-2.1)		-0.03 (-2.7)		-0.02 (-1.4)		-0.03 (-2.6)			
Elasticity (at the average limit price)		-0.02 (-3.2)		-0.02 (-0.6)		-0.05 (-2.5)		-0.05 (-1.6)		
Range size			0.60 (1.4)	0.66 (1.2)	0.77 (2.3)	0.64 (1.9)	1.26 (2.4)	1.32 (2.3)		
Adjusted <i>R</i> -squared	3.0%	5.4%	7.3%	-0.7%	12.5%	17.3%	26.5%	25.6%		
<i>N</i>	36	34	36	34	36	34	36	34		

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Table IX
Aftermarket Return by Bidder Types: IPOs

This table reports regression coefficients (and heteroskedasticity-adjusted *t*-statistics in parentheses) for various model specifications. The dependent variable is the first-day aftermarket return benchmarked relative to the domestic stock market index. The average limit price is the quantity-weighted average of all limit prices. *Oversubscription* is the logarithm of 1 + total demand/supply of shares. Large (small) bids are bids with a quantity above (below) the median in an issue. Frequent (infrequent) bidders are bidders who participate in at least (fewer than) three issues. Early bids are bids received at least three days prior to the closing of the book. Late bids are bids received in the last three days. The symbols ^{†††}, ^{††} denote *pairs of coefficients* that are significantly different from each other at the one and five percent confidence levels, respectively.

Dependent Variable: First-Day Return (Benchmarked)	Reg 1	Reg 2	Reg 3
Intercept	-0.10 (-2.6)	-0.05 (-1.7)	-0.10 (-1.6)
Percentage difference between average limit price and issue price	-0.47 (-1.2)	-0.37 (-1.0)	0.18 (0.3)
Oversubscription (large bids)	0.13 ^{††} (3.6)		
Oversubscription (small bids)	-0.20 ^{†††} (-2.5)		
Oversubscription (frequent bidders)		0.08 ^{††} (3.4)	
Oversubscription (infrequent bidders)		-0.02 ^{††} (-0.7)	
Oversubscription (early bids)			0.04 (1.7)
Oversubscription (late bids)			0.09 (2.2)
Adjusted <i>R</i> -squared	43.6%	42.9%	43.6%
<i>N</i>	34	34	34

6.10 Table IX: Aftermarket Return by Bidder Types: IPOs

Read:

- Oversubscription from large bids predicts first-day returns positively, while oversubscription from small bids is negative.
- Oversubscription from frequent bidders predicts returns more strongly than oversubscription from infrequent bidders.
- Late oversubscription is more informative than early oversubscription.

Interpretation: The same bidder types that influence offer pricing also contain residual aftermarket information. The bank is right to pay attention to large and frequent bidders, but it still does not fully incorporate their demand into price.

Economic message: Investor identity matters because some investors are more informative than others.

6.11 Table X: The Choice of the Issue Price (SEOs)

Read:

- Average limit prices strongly influence SEO issue prices, even though an observable premarket price exists.
- Oversubscription and elasticity are less important for SEO pricing.
- Large, frequent, and favored bidders again have more influence on issue pricing.

Interpretation: The SEO evidence is important because the underwriter already observes a market price. The book still adds information because the outstanding shares may be illiquid and the new issue may be large relative to existing float.

Economic message: Bookbuilding generates useful information even in markets with preexisting prices.

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Table X
The Choice of the Issue Price (SEOs)

This table reports regression coefficients (and heteroskedasticity-adjusted *t*-statistics in parentheses) for various model specifications. The dependent variable in Regressions 1 to 4 is the issue price normalized by the premarket price, while in Regression 5 the dependent variable is the absolute value of the percentage difference between issue price and average limit price. *Average limit price* is the quantity-weighted average of all limit prices and is normalized by the premarket price. *Oversubscription* is the logarithm of 1 + total demand/supply of shares. *Elasticity* is computed from the average limit price to a price one percent higher. Large (small) bids are bids with a quantity above (below) the median in an issue. Frequent (infrequent) bidders are bidders who participate in at least (fewer than) three issues. Favored bids are bids that are awarded more shares (as a percentage of the bid quantity) than the median bidder. The symbol [†] denotes pairs of coefficients that are significantly different from each other at the 10 percent confidence level.

	Dependent Variable: Issue Price (Normalized by Premarket Price)				Dependent Variable: Absolute Value of the Percentage Difference between Issue Price and Average Limit Price
	Reg 1	Reg 2	Reg 3	Reg 4	Reg 5
Intercept	-0.01 (-1.3)	-0.02 (-1.9)	-0.02 (-1.7)	-0.02 (-2.2)	0.01 (2.1)
Average limit price (all bids)	0.58 (5.6)				
Oversubscription (all bids)	0.00 (0.1)	0.00 (0.5)	0.00 (0.6)	0.00 (0.3)	0.00 (0.7)
Elasticity (at the average limit price)	-0.04 (-0.7)				-0.13 (-2.7)
Average limit price (large bids)		0.64 [†] (2.6)			
Average limit price (small bids)		0.09 [†] (0.9)			
Average limit price (frequent bidders)			0.38 (2.2)		
Average limit price (infrequent bidders)			0.40 (5.0)		
Average limit price (favored bids)				0.75 [†] (5.0)	
Average limit price (nonfavored bids)				-0.01 [†] (-0.1)	
Adjusted <i>R</i> -squared	55.6%	57.2%	59.4%	64.3%	2.4%

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Table XI
Aftermarket Return (SEOs)

This table reports regression coefficients (and heteroskedasticity adjusted *t*-statistics in parentheses) for various model specifications. The dependent variable is the first-day aftermarket return benchmarked relative to the domestic stock market index. *Oversubscription* is the logarithm of 1 + total demand/supply of shares. The average limit price is the quantity-weighted average of all limit prices. *Elasticity* is computed from the issue price to a price one percent higher. *Preissue volatility* is based on market prices during the one month before bookbuilding. Large (small) bids are bids with a quantity above (below) the median in an issue. Frequent (infrequent) bidders are bidders who participate in at least (fewer than) three issues. Early bids are bids received at least three days prior to the closing of the book. Late bids are bids received in the last three days. The symbols ^{†††} denote pairs of coefficients that are significantly different from each other at the one and five percent confidence levels, respectively.

Dependent Variable: First-Day Return (Benchmarked)	Reg 1	Reg 2	Reg 3	Reg 4	Reg 5
Intercept	-0.03 (-0.9)	-0.03 (-1.3)	-0.04 (-1.9)	-0.04 (-2.3)	-0.03 (-1.3)
Percentage difference between premarket price and issue price	0.96 (2.2)	0.93 (2.0)	0.99 (2.4)	1.20 (3.6)	1.06 (2.5)
Oversubscription (all bids)	0.03 (1.3)	0.02 (0.9)			
Percentage difference between average limit price and issue price	0.31 (0.6)	0.03 (0.1)	0.18 (0.4)	0.09 (0.2)	0.13 (0.3)
Elasticity (at the issue price)	0.002 (2.2)				
Preissue volatility		0.09 (1.7)			
Oversubscription (large bids)			0.06 ^{††} (2.7)		
Oversubscription (small bids)			-0.12 ^{††} (-1.7)		
Oversubscription (frequent bidders)				0.06 ^{†††} (5.0)	
Oversubscription (infrequent bidders)				-0.04 ^{†††} (-2.3)	
Oversubscription (early bids)					0.01 (0.7)
Oversubscription (late bids)					0.04 (2.0)
Adjusted <i>R</i> -squared	18.9%	25.8%	28.2%	39.1%	19.1%
<i>N</i>	25	25	25	25	24

6.12 Table XI: Aftermarket Return (SEOs)

Read:

- When the issue price deviates from the premarket price, the aftermarket price tends to revert toward the premarket price.
- The coefficient on the premarket-price gap is close to one in several specifications.

- Oversubscription and elasticity are not consistently significant for SEO aftermarket returns.

Interpretation: SEO aftermarket behavior is dominated by reversion toward the premarket price. The order book matters for pricing, but the preexisting market price anchors aftermarket valuation.

Economic message: SEO bookbuilding can adjust issuance prices around a noisy market benchmark, but the market benchmark remains powerful.

7 Interpretive synthesis

- **Bookbuilding as information extraction.** The strongest fact is that average limit prices explain most of the variation in normalized IPO issue prices.
- **Price information dominates quantity information.** Oversubscription matters, but limit prices have greater explanatory power because they directly reveal valuations.
- **Relationships and allocations matter.** Frequent, large, and favored bidders provide more influential information, consistent with repeated-game incentives and allocation rewards.
- **Partial adjustment is not a refutation.** Residual predictability in aftermarket returns can reflect optimal compensation for information revelation.
- **Public information is mediated.** Market and industry signals affect bids; underwriters rely on investors' interpretation rather than directly using public variables mechanically.
- **SEOs show bookbuilding beyond IPO uncertainty.** Even with a premarket price, underwriters use the book when liquidity or issue size makes the market benchmark imperfect.

8 Short summary

- The paper studies 63 order books from a major European investment bank.
- The sample includes 37 IPOs and 26 SEOs from 1995–1999.
- IPO issue prices are normalized by the initial price range; SEO issue prices are normalized by the premarket price.
- Average limit prices explain most of the cross-sectional variation in normalized IPO issue prices.
- Oversubscription matters, but less than limit prices.
- Large, frequent, and allocation-favored investors provide the most influential bids.
- Public information affects issue prices mainly through bid reactions.
- Some information in the book predicts first-day aftermarket returns, implying partial adjustment.
- Demand elasticity predicts aftermarket consensus and volatility.
- SEO prices are also influenced by limit prices, even though a premarket price exists.

9 Conclusion

9.1 Core conclusion

Cornelli and Goldreich provide direct evidence that bookbuilding is informative. The underwriter uses institutional bids, especially limit prices from large, frequent, and favored bidders, to choose the issue price.

9.2 Economic intuition

Limit bids reveal how much informed investors are willing to pay. The underwriter rewards information revelation with allocations and underpricing, but does not fully incorporate all information into the price because investors must be compensated for truthful revelation.

9.3 Incremental contribution in one sentence

The paper's distinctive contribution is to open the black box of the bookbuilding process by using actual confidential order books to show exactly which bidder information affects issue prices and aftermarket outcomes.

9.4 Takeaway

Bookbuilding should be viewed as a market-design institution for eliciting information from investors, not merely as a placement technology for distributing shares.